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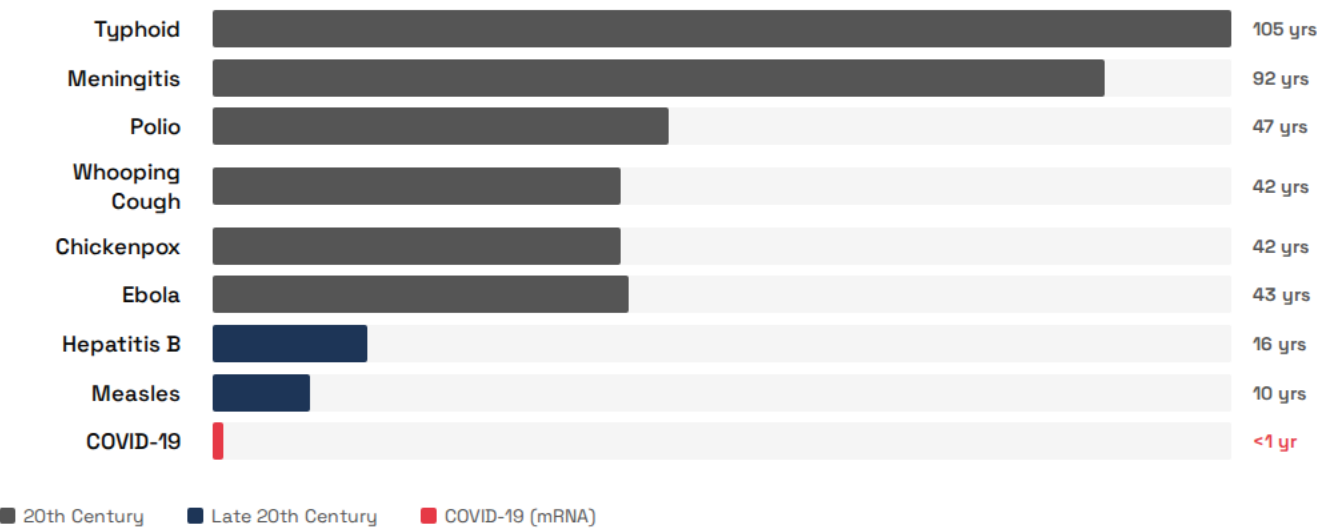
# The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time — A  
Data-Driven Overview, 1796–2020

Source: Our World in Data



# From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



# 100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

| DISEASE      | PRE-VACCINE CASES | POST-VACCINE CASES | CASE REDUCTION | DEATH REDUCTION |
|--------------|-------------------|--------------------|----------------|-----------------|
| Diphtheria   | 158 / million     | 0                  | 100%           | 100%            |
| Smallpox     | 250 / million     | 0                  | 100%           | 100%            |
| Acute Polio  | 141 / million     | 0                  | 100%           | 100%            |
| Measles      | 3,044 / million   | 0.2 / million      | 99.99%         | 100%            |
| Rubella      | 242 / million     | 0.04 / million     | 99.98%         | 100%            |
| Pertussis    | 1,534 / million   | 52 / million       | 96.6%          | 99.7%           |
| Hepatitis A  | 465 / million     | 51 / million       | 89%            | 88.7%           |
| Hepatitis B  | 273 / million     | 44 / million       | 83.9%          | 83.6%           |
| Pneumococcal | 233 / million     | 139 / million      | 40.5%          | 31.3%           |

Data: Roush and Murphy (2007). Rates per million population per year. Highlighted rows = 100% elimination.



# Measles Cases Collapsed Across All 50 US States After Vaccine Mandates



Each cell represents a US state. Color intensity shows measles cases per 100,000 (1929–2022). Red = high, green = eliminated.

**1963** **First Measles Vaccine**  
Developed by John Enders. Before this, nearly every state experienced high incidence (100–1,000+ cases per 100,000).

**1971** **MMR Vaccine Introduced**  
Maurice Hilleman combines measles, mumps, and rubella into a single shot, streamlining childhood immunization.

**1980** **Mandatory Kindergarten Vaccination**  
Cases drop to near zero across all states. Only sporadic outbreaks remain.



# Smallpox: The Only Human Disease Eradicated Through Vaccination



**Human-Only Host**

The virus infected only humans — no animal reservoir to sustain transmission.

**Visible Symptoms**

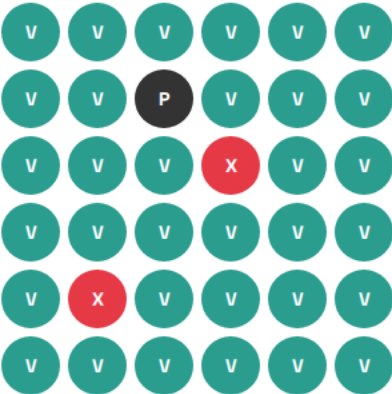
Distinctive rash made cases easy to identify and isolate quickly.

**Cheap, Effective Vaccine**

The vaccine was inexpensive, stable, and conferred long-lasting immunity.



# Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



● Vaccinated (protected) ● Vulnerable (shielded) ● Pathogen (contained)

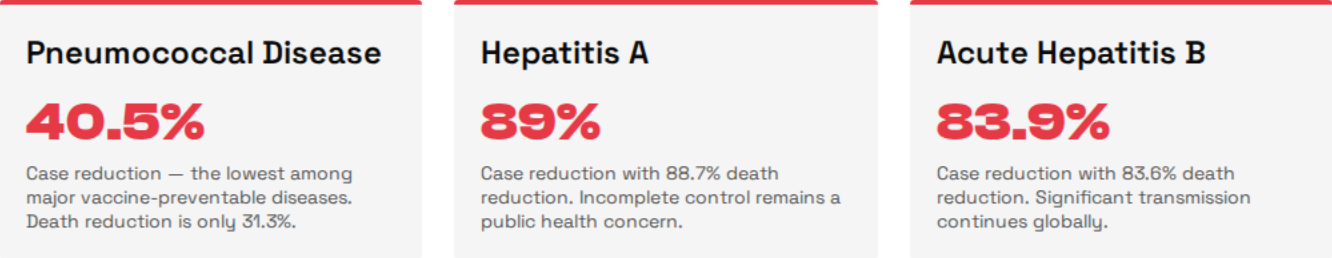
**Vaccines protect entire communities, not just individuals.**

When enough people are vaccinated, pathogens cannot spread effectively. The chain of transmission is broken.

This shields those who **cannot be vaccinated** — newborns, immunocompromised individuals, and cancer patients on chemotherapy.



# Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



## Key Barriers to Full Coverage

- Uneven global access
- Supply shortages
- Vaccine hesitancy
- Poor scientific understanding
- Trust deficits



## Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- 1** **100% elimination achieved** for diphtheria, smallpox, and acute polio in the US — cases and deaths reduced to zero.
- 2** **Development timelines have collapsed** from over 100 years (typhoid, 1884–1989) to under 1 year (COVID-19, 2020), driven by mRNA technology and genomic tools.
- 3** **Measles cases dropped to near zero** across all 50 US states following the 1980 kindergarten vaccination mandate.
- 4** **Smallpox remains the only human disease eradicated** through vaccination, certified in 1980 after a 21-year global campaign.
- 5** **Coverage gaps persist** for pneumococcal disease (40.5%), hepatitis A (89%), and hepatitis B (83.9%), highlighting the need for improved access and newer vaccines.

**Call to action:** Global health systems must prioritize equitable access, strengthen supply chains, and rebuild public trust in vaccination to close remaining coverage gaps and prepare for the next pandemic.

